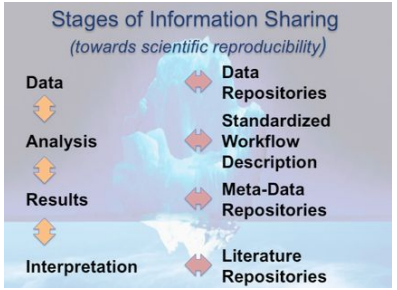
Best Practices for Biomedical Research Data Management

Module 9: Scientific Research Team Research Project Overview

Dr. David Kennedy, PhD, UMass Medical School



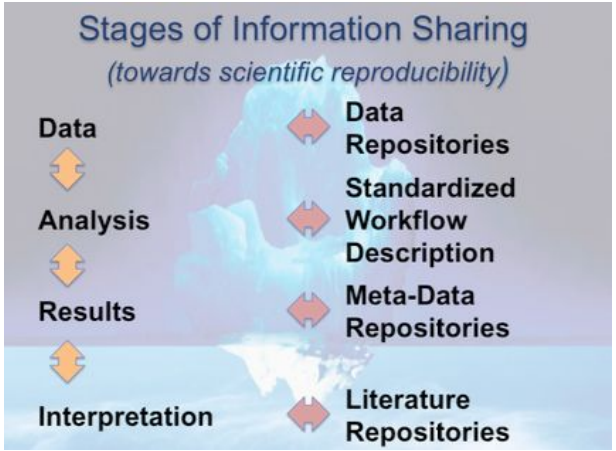
This video will provide an overview of the 'CANDIShare' Grant elaborated on in the videos included in Module 9. The project aims to expand Neuroimaging data sharing within the Child Psychiatry community; create a portal to integrate resources and enhance data discovery within the community; and perform novel investigation into the neurobiological underpinnings of child psychiatric disorders.

CC BY NC SA

David N. Kennedy, PhD., Leah Honor, MLIS & Christian Haselgrove

Child and Adolescent NeuroDevelopment Initiative

Lamar Soutter Library

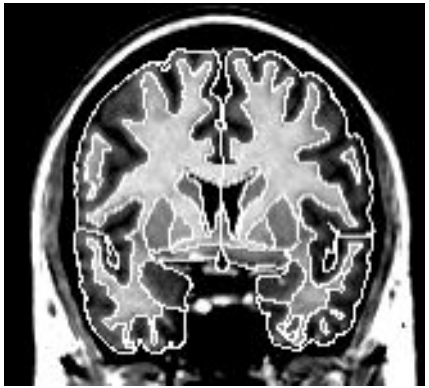


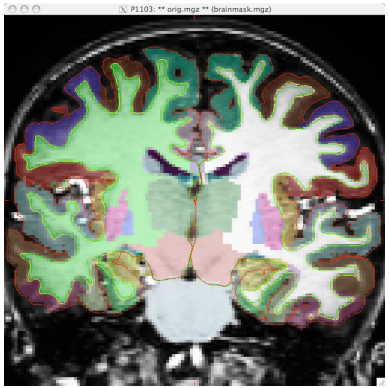
University of Massachusetts Medical School UMASS

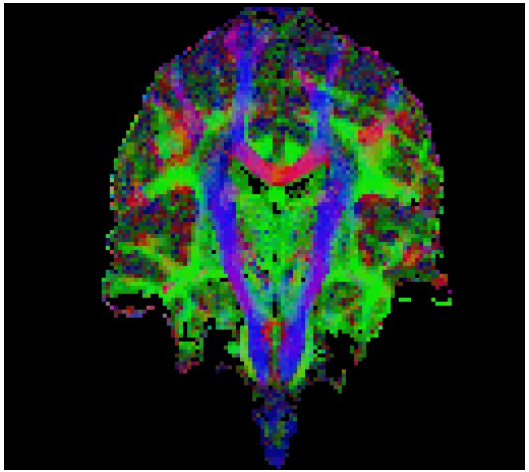
CANDI

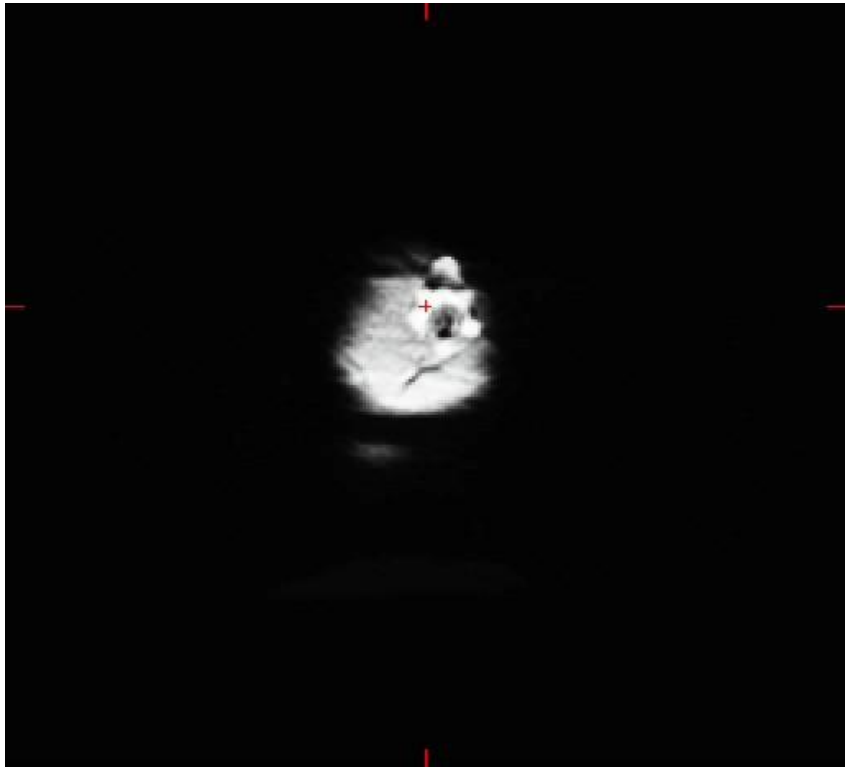
NITRC

What Data?



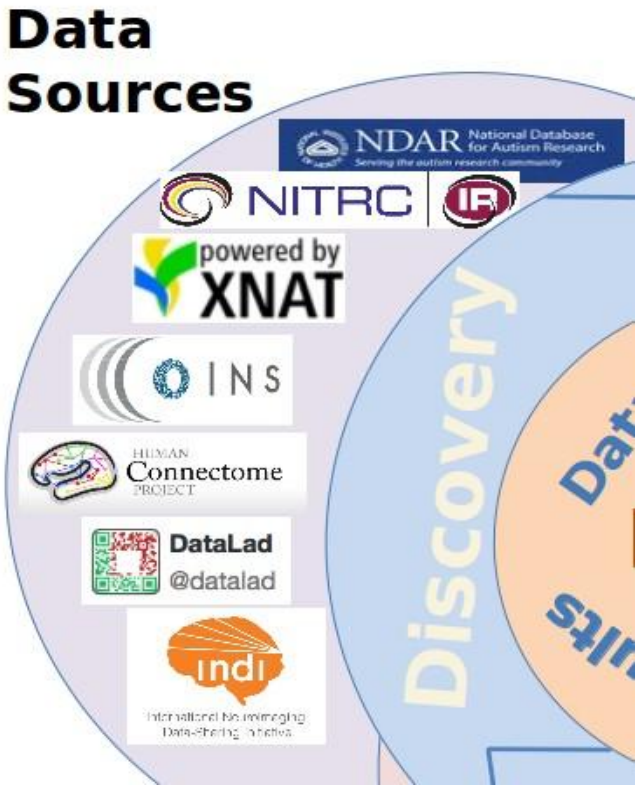






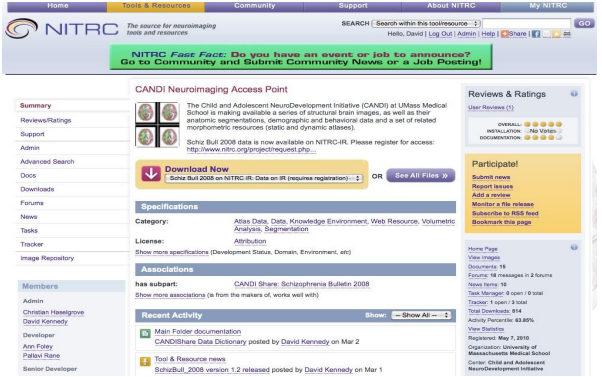
Data Sharing Happens

(and is becoming more and more required…)



Integrated Publication

http://www.nitrc.org/projects/candi_share/



Schizophrenia Bulletin vol. 34 no. 1 pp. 37–46, 2008
doi:10.1093/schbul/sbm120
Advance Access publication on November 13, 2007

Diagnostic and Sex Effects on Limbic Volumes in Early-Onset Bipolar Disorder and Schizophrenia

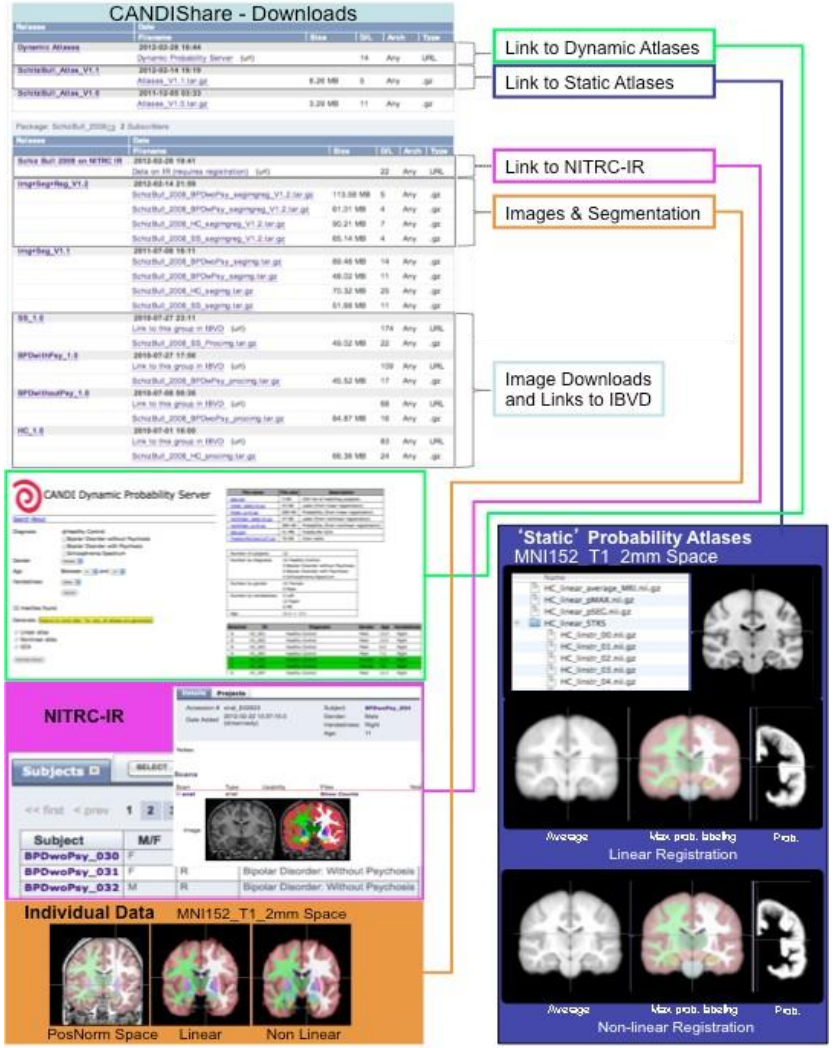
Jean A. Frazier[1–4], Steven M. Hodge[5], Janis L. Breeze[2,3], Anthony J. Giuliano[2,3,6], Janine E. Terry[3], Constance M. Moore[2,3,7], David N. Kennedy[3,5,8], Melissa P. Lopez-Larson[2–4], Verne S. Caviness[3,8,9], Larry J. Seidman[2,6,10], Benjamin Zablotsky[3], and Nikos Makris[5,11]

[2]Department of Psychiatry, Harvard Medical School; [3]Child and Adolescent Neuropsychiatric Research Program, Cambridge Health Alliance; [4]Department of Psychiatry, McLean Hospital; [5]Center for Morphometric Analysis, Massachusetts General Hospital; [6]Department of Psychiatry, Massachusetts Mental Health Center at the Beth Israel Medical Center; [7]Brain Imaging Center, McLean Hospital; [8]Department of Neurology, Harvard Medical School; [9]Department of Pediatric Neurology, Massachusetts General Hospital; [10]Department of Psychiatry,

hormone receptors. In addition, smaller thalamus was associated with SZ while larger right NA volumes were most related to BPD. This study underscores the importance of assessing diagnostic effects and sex effects on the brain in future studies and provides evidence that boys and girls with SZ and BPD may have differential patterns of neuropathology associated with disease expression.

Key words: mood disorders/psychosis/brain imaging technique/child psychiatry

Introduction



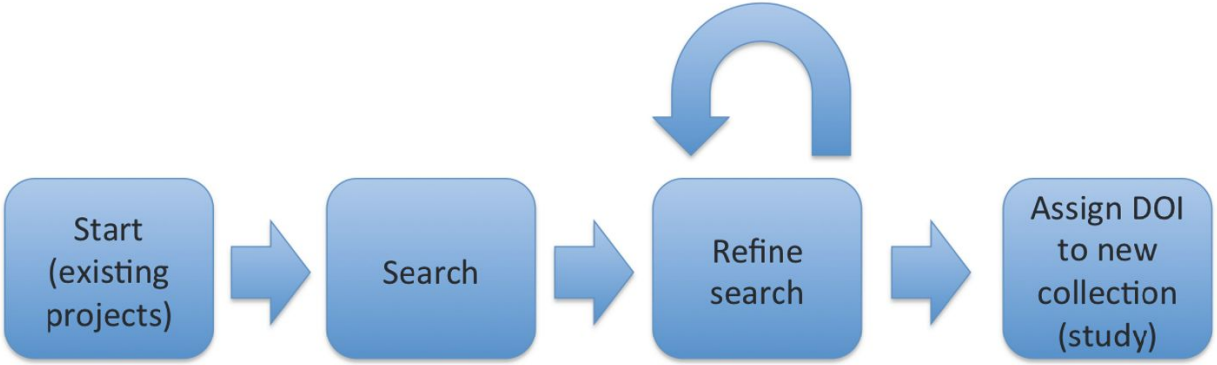
With great sharing…
Comes the need for great credit…

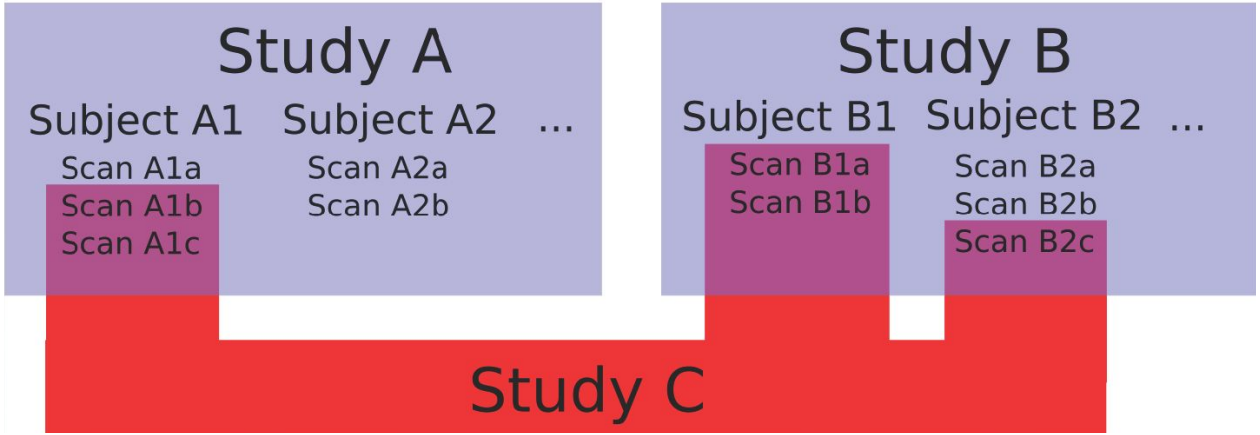
- How to assign credit has been talked about by lots of people for a long time, but nothing was really being done about it
- Neuroimaging is a somewhat ‘special case’. Standards, ubiquity, repositories, etc.

Neuroimaging: How to cite?

- Assign DOIs to the finest independently sharable object
  - Images
  - Which can be grouped into Collections
    - Projects
    - Functional Collections

- Reference Implementation
  - http://iaf.virtualbrain.org
    - XNAT
    - https://github.com/chaselgrove/ezid and https://github.com/chaselgrove/doi


Start (existing projects)
Search
Refine search
Assign DOI to new collection (study)


Study A
Subject A1
Subject A2
...
Scan A1a
Scan A1b
Scan A1c
Scan A2a
Scan A2b
Study B
Subject B1
Subject B2
...
Scan B1a
Scan B1b
Scan B2a
Scan B2b
Scan B2c
Study C

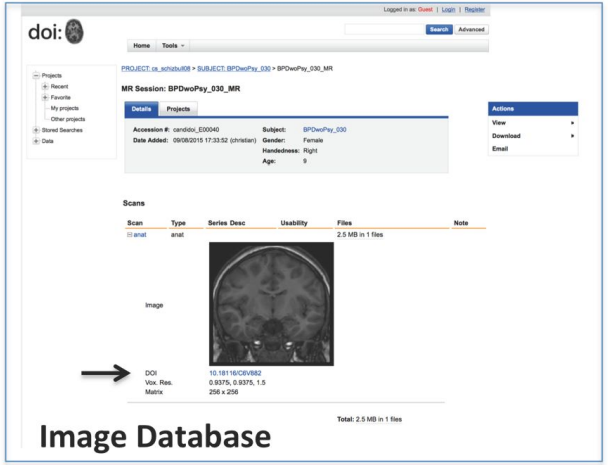


Image Database

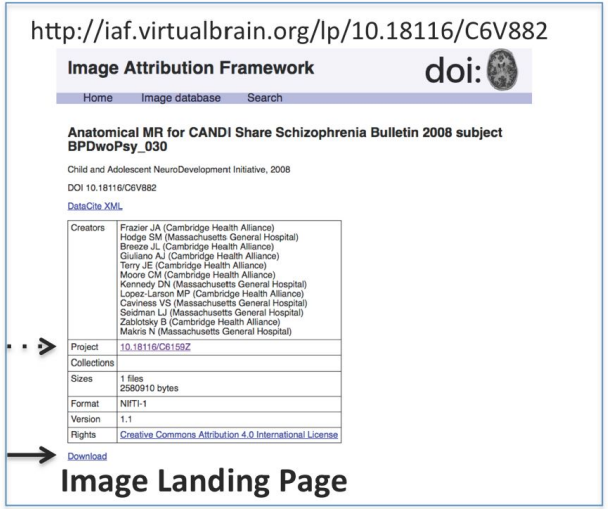
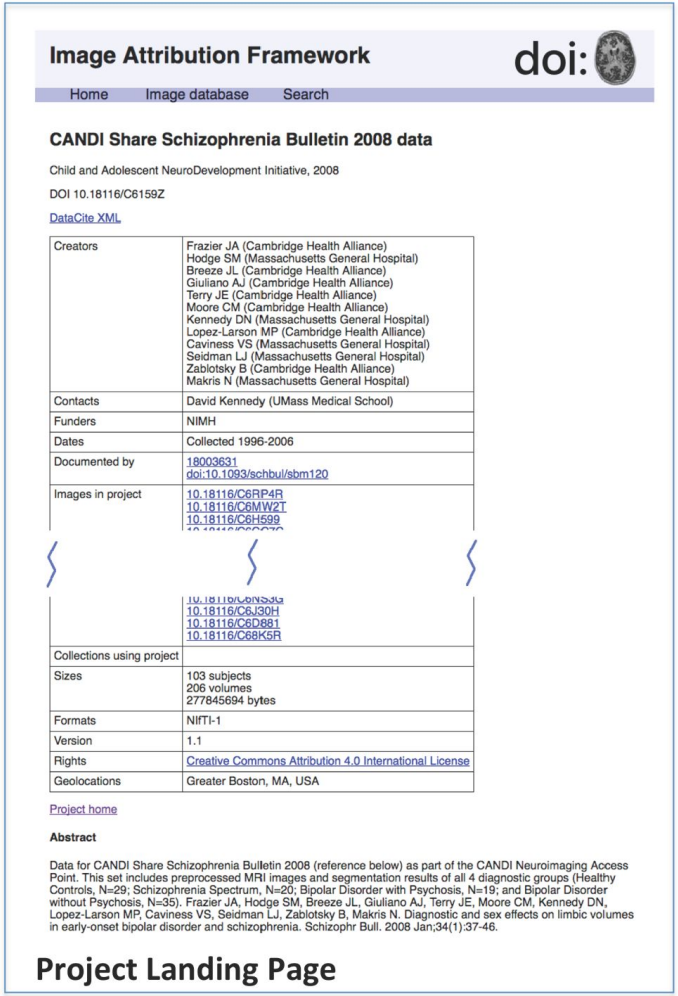


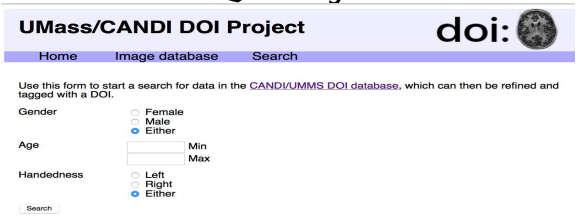
Image Landing Page



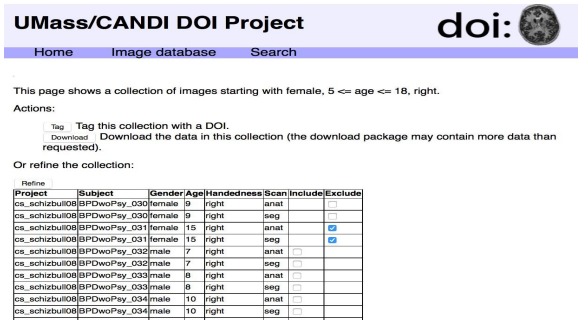
Project Landing Page

Functional DOI Generation

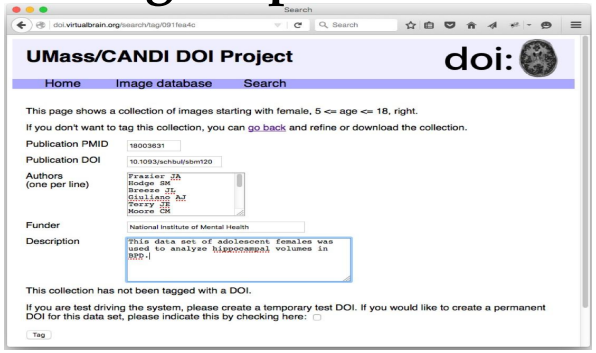
Query



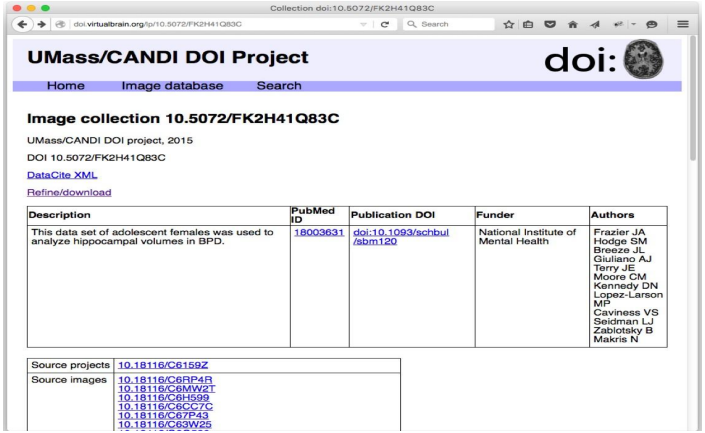
Result/Selection



Tag Preparation



New Functional>Level Collection Landing Page

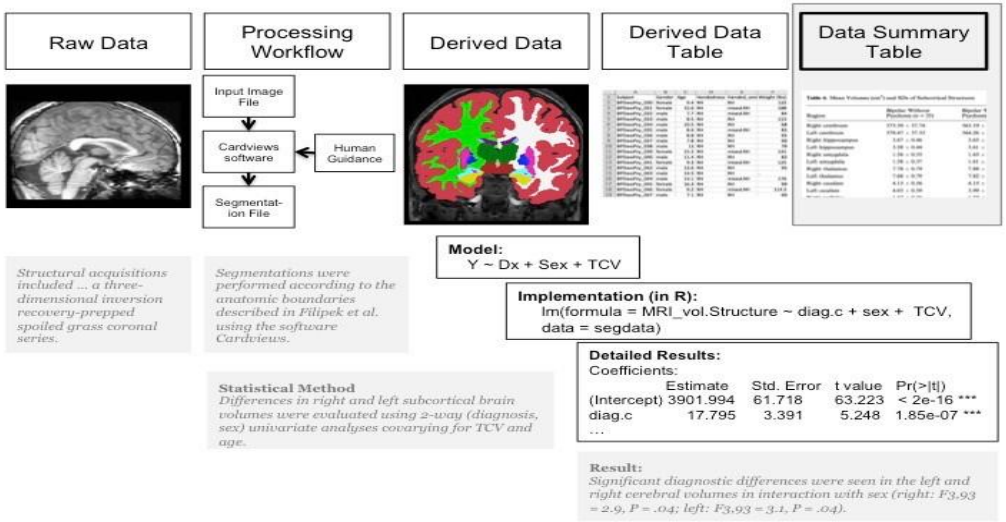


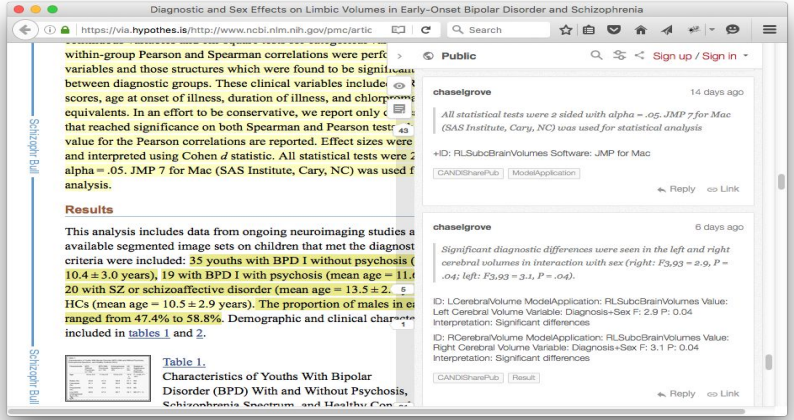
Result

- Manuscript text and citations like:
  - Objective 2, attribution, is also facilitated by the use of the DOI, at both individual (Frazier et al. 2008a) and collective levels (project: Frazier et al. 2008b; functioanl collection: Breeze, et al, 2016), through the pre-existing utilities designed to track publication citations. One of the main benefits for using DOIs explicitly is the existing infrastructure for monitoring, tracking and indexing their use.
- Can map to citations like:
  - Breeze, JL, Caplan, D, Caviness, VS, Frazier, JA, Giuliano, AJ, Haselgrove, C, Hodge, SM, Kennedy, DN, Lopez-Larson, MP, Makris, N, Moore, CM, Seidman, LJ, Terry, JE, Worth, AJ, Zablotsky, B. 2016. Image collection. UMass/CANDI Image Attribution Framework. http://dx.doi.org/10.18116/C6H02X.
  - Frazier, JA et al. 2008a. “Anatomical MR for CANDI Share Schizophrenia Bulletin 2008 Subject BPDwoPsy_057.” *Child and Adolescent NeuroDevelopment Initiative*. http://dx.doi.org/10.18116/C6RP4R.
  - Frazier, JA et al. 2008b. “CANDI Share Schizophrenia Bulletin 2008 Data.” *Child and Adolescent NeuroDevelopment Initiative*. http://dx.doi.org/10.18116/C6159Z.
- Which can then be tracked with the ‘usual’ DOI-based tracking methods.

In Progress

- Data Citation Reporting and Impact Calculation
- Extracting even MORE our of the current literature (exposing the ‘dark matter’ of the bibliome…)





Summary

- We have presented a system for neuroimaging data citation and credit with a practical implementation that meets the objectives of unique identification, data use tracking, and integration with traditional credit attribution systems
- Key topical areas covered:
    - Metadata Schema
    - Landing Pages
    - Researcher Perspective
    - Data Host Perspective
    - Cost implications
    - Alternative Approaches
    - Barriers to wide-spread adoption within the neuroimaging community
        - Researchers
        - Data Hosting Sites
        - Journals (editors, reviewers, publishers)
        - Funding Agencies